1.4 Constraints

Every physical system has a particular number of *degrees of freedom.* The number of degrees of freedom is the number of independent coordinates needed to completely specify the position of every part of the system. To describe the position of a free particle one must specify the values of three coordinates (say, x, y and z). Thus a free particle has three degrees of freedom. For a system of two free particles you need to specify the positions of both particles. Each particle has three degrees of freedom, so the system as a whole has six degrees of freedom. In general a mechanical system consisting of N free particles will have $3N$ degrees of freedom.

When a system is acted upon by forces of constraint, it is often possible to reduce the number of coordinates required to describe the motion. Thus, for example, the motion of a particle on a table can be described in terms of x and y, the constraint being that the z coordinate (defined to be perpendicular to the table) is given by $z =$ constant. The motion of a particle on the surface of a sphere can be described in terms of two angles, and the constraint itself can be expressed by $r =$ constant. *Each constraint reduces by one the number of degrees of freedom.*

In many problems the system is constrained in some way. Examples are a marble rolling on the surface of a table or a bead sliding along a wire. In these problems the particle is not completely free. There are forces acting on it which restrict its motion. If a hockey puck is sliding on smooth ice, the gravitational force is acting downward and the normal force is acting upward. If the puck leaves the surface of the ice, the normal force ceases to act and the gravitational force quickly brings it back down to the surface. At the surface, the normal force prevents the puck from continuing to move downward in the vertical direction. The force exerted by the surface on the particle is called a *force of constraint*.

If a system of N particles is acted upon by k constraints, then the number of generalized coordinates needed to describe the motion is $3N - k$. (The number of Cartesian coordinates is always $3N$, but the number of (independent) generalized coordinates is $3N - k$.)

A constraint is a relationship between the coordinates. For example, if a particle is constrained to the surface of the paraboloid formed by rotating a parabola about the z axis, then the coordinates of the particle are related by $z - x^2/a - y^2/b = 0$. Similarly, a particle constrained to the surface of a sphere has coordinates that are related by $x^2 + y^2 + z^2 - a^2 = 0$.

If the equation of constraint (the relationship between the coordinates) can be expressed in the form

$$f(q_1, q_2, \ldots, q_n, t) = 0, \tag{1.8}$$

then the constraint is called *holonomic*.[1]

The key elements in the definition of a holonomic constraint are: (1) the equals sign and (2) it is a relation involving the *coordinates*. For example, a possible constraint is that a particle is always outside of a sphere of radius a. This constraint could be expressed as $r \geq a$. This is *not* a holonomic constraint. Sometimes a constraint involves not just coordinates but also velocities or differentials of the coordinates. Such constraints are also not holonomic.

As an example of a non-holonomic constraint consider a marble rolling on a perfectly rough table. The marble requires five coordinates to completely describe its position and orientation, two linear coordinates to give its position on the table top and three angle coordinates to describe its orientation. If the table top is perfectly smooth, the marble can *slip* and there is no relationship between the linear and angular coordinates. If, however, the table top is rough there are relations (constraints) between the angle coordinates and the linear coordinates. These constraints will have the general form $dr = a d\theta$, which is a relation between differentials. If such differential expressions can be integrated, then the constraint becomes a relation between coordinates, and it is holonomic. But, in general, rolling on the surface of a plane does not lead to integrable relations. That is, in general, the *rolling* constraint is not holonomic because the rolling constraint is a relationship between *differentials*. The equation of constraint does not involve *only* coordinates. (But rolling in a straight line is integrable and hence holonomic.)

A holonomic constraint is an equation of the form of (1.8) relating the generalized coordinates, and it can be used to express one coordinate in terms of the others, thus reducing the number of coordinates required to describe the motion.

Exercise 1.5 A bead slides on a wire that is moving through space in a complicated way. Is the constraint holonomic? Is it scleronomous?

[1] A constraint that does not contain the time explicitly is called *scleronomous*. Thus $x^2 + y^2 + z^2 - a^2 = 0$ is both holonomic and scleronomous.

Exercise 1.6 Draw a picture showing that a marble can roll on a tabletop and return to its initial position with a different orientation, but that there is a one-to-one relation between angle and position for rolling in a straight line.

Exercise 1.7 Express the constraint for a particle moving on the surface of an ellipsoid.

Exercise 1.8 Consider a diatomic molecule. Assume the atoms are point masses. The molecule can rotate and it can vibrate along the line joining the atoms. How many rotation axes does it have? (We do not include operations in which the final state cannot be distinguished from the initial state.) How many degrees of freedom does the molecule have? (Answer: 6.)

Exercise 1.9 At room temperature the vibrational degrees of freedom of O_2 are "frozen out." How many degrees of freedom does the oxygen molecule have at room temperature? (Answer: 5.)

1.5 Virtual displacements

A *virtual displacement*, δx_i, is defined as an infinitesimal, instantaneous displacement of the coordinate x_i, consistent with any constraints acting on the system.

To appreciate the difference between an ordinary displacement dx_i and a virtual displacement δx_i, consider the transformation equations

$$x_i = x_i(q_1, q_2, \ldots, q_n, t) \qquad i = 1, 2, \ldots, 3N.$$

Taking the differential of the transformation equations we obtain

$$\begin{aligned} dx_i &= \frac{\partial x_i}{\partial q_1} dq_1 + \frac{\partial x_i}{\partial q_2} dq_2 + \cdots + \frac{\partial x_i}{\partial q_n} dq_n + \frac{\partial x_i}{\partial t} dt \\ &= \sum_{\alpha=1}^{n} \frac{\partial x_i}{\partial q_\alpha} dq_\alpha + \frac{\partial x_i}{\partial t} dt. \end{aligned}$$

But for a virtual displacement δx_i, time is frozen, so

$$\delta x_i = \sum_{\alpha=1}^{n} \frac{\partial x_i}{\partial q_\alpha} dq_\alpha. \tag{1.9}$$

1.6 Virtual work and generalized force

Suppose a system is subjected to a number of applied forces. Let F_i be the force component along x_i. (Thus, for a two-particle system, F_5 would be the force in the y direction acting on particle 2.) Allowing all the Cartesian coordinates to undergo virtual displacements δx_i, the *virtual work* performed by the applied forces will be

$$\delta W = \sum_{i=1}^{3N} F_i \delta x_i .$$

Substituting for δx_i from Equation (1.9)

$$\delta W = \sum_{i=1}^{3N} F_i \left(\sum_{\alpha=1}^{3N} \frac{\partial x_i}{\partial q_\alpha} \delta q_\alpha \right).$$

Interchanging the order of the summations

$$\delta W = \sum_{\alpha=1}^{3N} \left(\sum_{i=1}^{3N} F_i \frac{\partial x_i}{\partial q_\alpha} \right) \delta q_\alpha .$$

Harkening back to the elementary concept of "work equals force times distance" we define the *generalized force* as

$$Q_\alpha = \sum_{i=1}^{3N} F_i \frac{\partial x_i}{\partial q_\alpha}. \tag{1.10}$$

Then the virtual work can be expressed as

$$\delta W = \sum_{\alpha=1}^{3N} Q_\alpha \delta q_\alpha .$$

Equation (1.10) is the definition of generalized force.

Example 1.1 *Show that the principle of virtual work ($\delta W = 0$ in equilibrium) implies that the sum of the torques acting on the body must be zero.*

Solution 1.1 *Select an axis of rotation in an arbitrary direction $\mathbf{\Omega}$. Let the body rotate about $\mathbf{\Omega}$ through an infinitesimal angle ϵ. The virtual displacement of a point P_i located at $\mathbf{R}_i$ relative to the axis is given by*

$$\delta \mathbf{R}_i = \epsilon \mathbf{\Omega} \times \mathbf{R}_i .$$

The work done by a force $\mathbf{F}_i$ acting on P_i is

$$\delta W_i = \mathbf{F}_i \cdot \delta \mathbf{R}_i = \mathbf{F}_i \cdot (\epsilon \mathbf{\Omega} \times \mathbf{R}_i) = \epsilon \mathbf{\Omega} \cdot (\mathbf{R}_i \times \mathbf{F}_i) = \epsilon \mathbf{\Omega} \cdot \mathbf{N}_i ,$$